



BAULKHAM HILLS HIGH SCHOOL

2012

YEAR 12 HALF-YEARLY

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 120 minutes
- Write using black or blue pen
- Board-approved calculators may be used
- All necessary working should be shown in every question
- Marks may be deducted for careless or badly arranged work
- Attempt all questions
- Start a new page for each question

Total marks – 70

Exam consists of 8 pages.

This paper consists of TWO sections.

Section 1 – Pages 3-5

Multiple Choice

Question 1-10 (10 marks)

Section 2 – Pages 6-9

Extended Response

Question 11- 14 (60 marks)

Standard integrals provided on page 10

Section I

10 marks

Attempt questions 1-10

Use the multiple choice answer sheet for question 1-10

1. Simplify $\frac{1}{i^9}$

- a) i b) $-i$ c) 1 d) -1

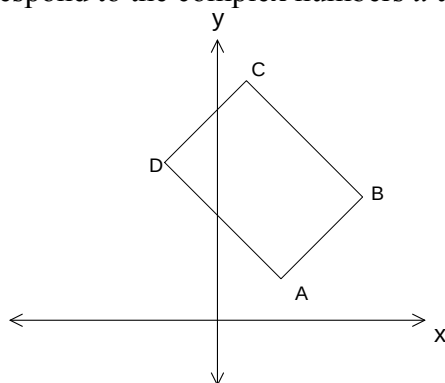
2. z^{-1} is equivalent to

- a) $\bar{z}$ b) $|z|^2$ c) $\frac{\bar{z}}{|z|}$ d) $\frac{\bar{z}}{|z|^2}$

3. $P(x) = x^3 + 8x^2 - ax + b$ where a and b are integers. If $1+3i$ is a root, which of the following is also a root?

- a) -10 b) -8 c) -3 d) 3

4. In the Argand diagram, ABCD is a rectangle and $AD = 2AB$. The vertices A and B correspond to the complex numbers u and v .



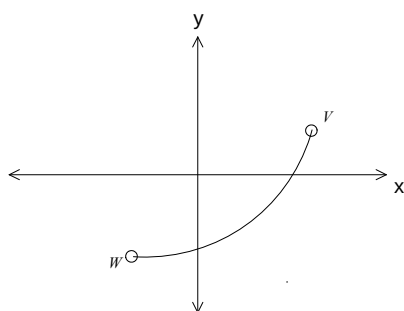
The point C is given by

- a) $u + 2iu$ b) $v + 2i(v - u)$ c) $v + 2iu$ d) $v + u + 2iu$

5. If $1, w, w^2$ are the cube roots of unity, simplify $w^7(1 + w)(-w^2 - 1)$

- a) $-w$ b) w c) w^2 d) $-w^2$

6. Which of the following could describe the locus of z shown in the diagram.



- a) $\arg\left(\frac{z - v}{z - w}\right) = \frac{\pi}{3}$ b) $\arg\left(\frac{z - v}{z - w}\right) = \frac{-\pi}{3}$
 c) $\arg\left(\frac{z - v}{z - w}\right) = \frac{2\pi}{3}$ d) $\arg\left(\frac{z - v}{z - w}\right) = \frac{-2\pi}{3}$

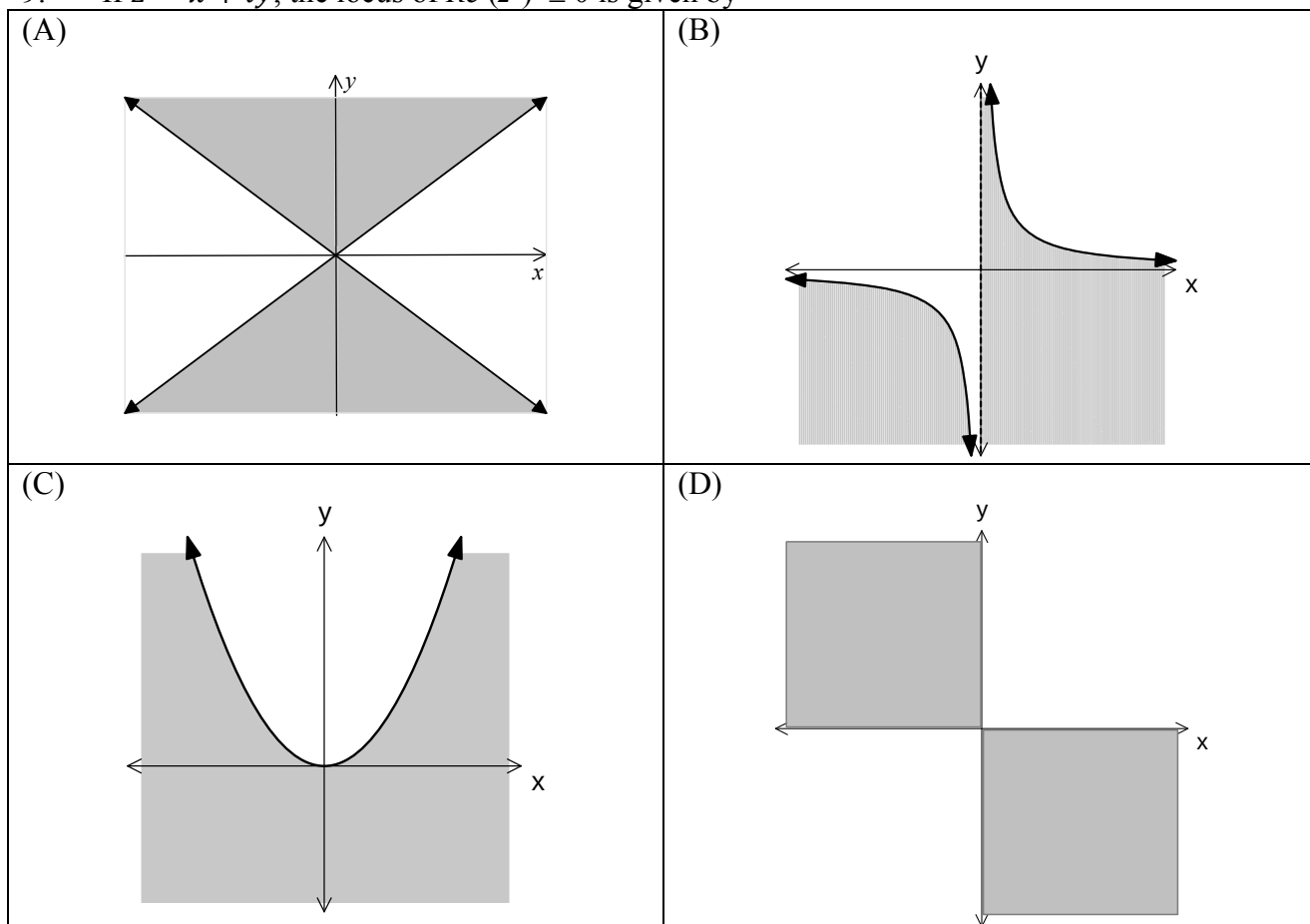
7. $z^2 - z^{-2}$ is equivalent to

- a) $2 \cos \theta$ b) $2 \cos 2\theta$ c) $2i \sin \theta$ d) $4i \sin \theta \cos \theta$

8. For $\frac{x^2}{16} + \frac{y^2}{25} = 1$, which of the following is true?

- a) foci are $\left(\pm \frac{12}{5}, 0\right)$ and directrices $x = \pm \frac{25}{3}$
 b) foci are $(0, \pm 3)$ and directrices $y = \pm \frac{20}{3}$
 c) foci are $\left(\pm \frac{12}{5}, 0\right)$ and directrices $x = \pm \frac{20}{3}$
 d) foci are $(0, \pm 3)$ and directrices $y = \pm \frac{25}{3}$

9. If $z = x + iy$, the locus of $\operatorname{Re}(z^2) \leq 0$ is given by



10. If $|z - (1 + 2i)| = 1$ the maximum value of $|z|$ is

- a) 2 b) 3 c) $1 + \sqrt{5}$ d) $2 + \sqrt{5}$

End of Section I

Section II – Extended Response

Attempt questions 11-14. Show all necessary working.

Answer each question on a SEPARATE PAGE Clearly indicate question number.

Each piece of paper must show your BOS number.

All necessary working should be shown in every question.

Question 11 (15 marks) – Start a new page

Marks

- a) The complex numbers

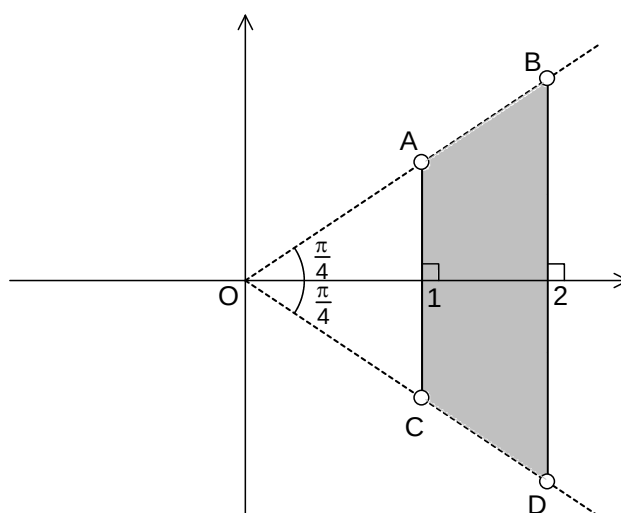
3

$$w = \frac{a}{1-2i} \text{ and } z = \frac{b}{2+i} \text{ (where } a \text{ and } b \text{ are real numbers) are such that } w + z = 1.$$

Find the values of a and b

- b) If $z = 3 + i\sqrt{3}$ find the smallest positive integer, n , for which $\text{Im}(z^n) < 0$ and the number is purely imaginary. 3

- c) i) Write down a pair of inequalities which describe the region in the Argand Diagram below. 2



- ii) The region in i) is rotated anticlockwise by $\frac{\pi}{4}$ radians to $A'B'C'D'$ 3

Find the complex number represented by the point of intersection of the diagonals of $A'B'C'D'$ in the form $z = a + ib$

- d) i) Show that the locus of z represented by $z\bar{z} + 10(z + \bar{z}) = 21$ is a circle centred at $(-10, 0)$ with radius 11. 2

- ii) Hence find which two purely imaginary numbers satisfy the equation $z\bar{z} + 10(z + \bar{z}) = 21$ 2

Question 12 (15 marks) – Start a new page**Marks**

- a) The roots of $x^3 + 6x^2 + 5x - 8 = 0$ are α, β and γ .

Find:

i) the value of $\alpha^2 + \beta^2 + \gamma^2$ **2**

ii) the value of $\alpha^3 + \beta^3 + \gamma^3$ **2**

iii) the monic polynomial with roots $\alpha^2, \beta^2, \gamma^2$ **3**

- b) i) Find the values of b and c in the polynomial **3**

$P(z) = z^2 + bz + c$ if b and c are real and $P(3 + i) = 0$

ii) Hence or otherwise, solve $z^3 - 7z^2 + 16z - 10 = 0$ over C . **2**

- c) The equation $8x^4 + 12x^3 - 30x^2 + 17x - 3 = 0$ has a root of multiplicity three. **3**
Solve the equation completely.

Question 13 (15 marks) – Start a new page

Marks

- a) α) Determine the real values of p for which

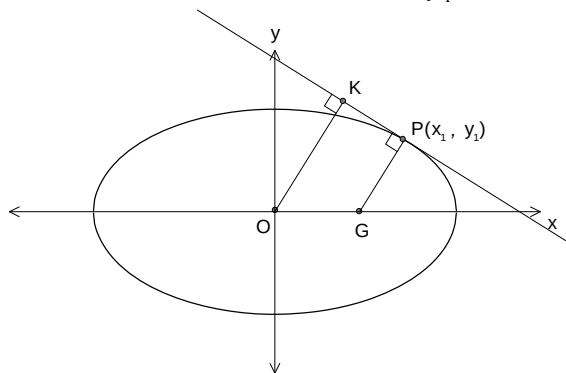
$$\frac{x^2}{4+p} + \frac{y^2}{9+p} = 1 \text{ defines}$$

- i) an ellipse 1
 ii) a hyperbola 2

β) Let $p = -5$ in the equation $\frac{x^2}{4+p} + \frac{y^2}{9+p} = 1$.

- i) Find the eccentricity 1
 ii) Find the coordinates of foci 1
 iii) Find the equations of the directrices 1
 iv) Find the equations of the asymptotes 1
 v) Hence draw a neat sketch showing all important features 1

- b) i) For the ellipse $x^2 + 4y^2 = 100$ show that the equation of the tangent at a point $P(x_1, y_1)$ on the curve is $y - y_1 = \frac{-x_1}{4y_1}(x - x_1)$ 2



- ii) If the normal at P meets the major axis at G show that the distance 3

$$PG = \frac{\sqrt{x_1^2 + 16y_1^2}}{4}$$

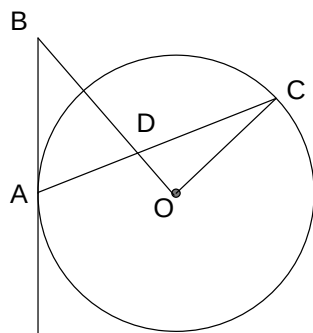
- iii) The perpendicular from the centre O to the tangent at P meets the tangent at K . Show that $PG \cdot OK$ is equal to the square of the length of the semi minor axis. 2

Question 14 (15 marks) – Start a new page

Marks

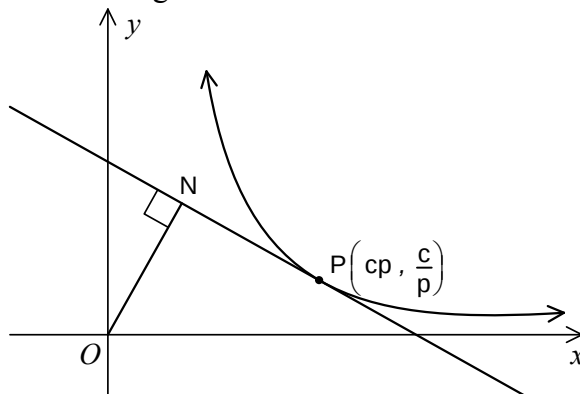
- a) AB is a tangent to a circle centred at O and AB=BD.

3



Prove that $\angle COB = 90^\circ$

- b) The line through O perpendicular to the tangent at $P\left(cp, \frac{c}{p}\right)$ on the rectangular hyperbola $xy = c^2$ meets the tangent at N



- i) Show that the coordinates of N are $\left(\frac{2cp}{1+p^4}, \frac{2cp^3}{1+p^4}\right)$ **3**
- ii) Find, in fully simplified form, the equation of the locus of N. **2**
- c) If $P(x)$ is divided by $(x-a)(x-b)$ a remainder of $R(x)$ is obtained. Show that the remainder is given by **3**
- $$R(x) = \left(\frac{P(a) - P(b)}{a - b}\right)x + \frac{aP(b) - bP(a)}{a - b}$$
- d) The points P, Q and R represent three different non zero complex numbers, p, q and r respectively. **4**

Show that these points must satisfy the condition $q^2 + r^2 + 2p^2 = 2(pq + pr)$ in order that $\triangle PQR$ forms a right angled isosceles triangle with $PQ=PR$.

End of Paper

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \quad \text{if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln(x + \sqrt{x^2 - a^2}), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln(x + \sqrt{x^2 + a^2})$$

NOTE: $\ln x = \log_e x$, $x > 0$

EXT 2 HALF YEARLY 2012

1. b
2. c
3. a
4. b
5. a
6. d
7. d
8. d
9. a
10. c

Q11a) $\frac{a}{1-i} + \frac{b}{2+i} = 1$

$$a(2+i) + b(1-i) = (1-i)(2+i) \quad \checkmark$$

$$2a+ib + i(a-ib) = 2-3i+2$$

$$2a+ib + i(a-ib) = 4-3i$$

$$2a+ib = 4 \quad (1) \quad \checkmark$$

$$a-ib = -3 \quad (2)$$

Q11b) $2a-4b = -6 \quad (3)$

Q11c) $5b = 6$

$$b = \frac{6}{5}$$

$$\therefore a = -3 + 4$$

$$a = 1$$

$$\therefore a = 1, b = \frac{6}{5}$$

11 b) $z = \sqrt{12} \operatorname{cis} \left(\frac{\pi}{6} \right)$ $\checkmark$

$$z^n = (\sqrt{12})^n \operatorname{cis} \left(\frac{n\pi}{6} \right)$$

$$\frac{n\pi}{6} = \frac{3\pi}{2}$$

$$n = \frac{18\pi}{2\pi}$$

$$n = 9 \quad \checkmark$$

11 c) i) $1 \leq \operatorname{Re}(z) \leq 2$ $\checkmark$
 $-\frac{\pi}{4} < \arg(z) < \frac{\pi}{4}$ $\checkmark$

ii) A is (1,1)

B is (2,-2)

Eqn of AB is $y-1 = \frac{-2-1}{2-1}(x-1)$

$$y-1 = -3(x-1)$$

$$y = -3x+4$$

By symmetry of shape diagonals intersect on x axis

when $y=0$

$$0 = -3x+4$$

$$3x = 4$$

$$x = \frac{4}{3}$$

$\therefore$ Pt intersection is $\left(\frac{4}{3}, 0 \right)$ $\checkmark$

when rotated pt of int given by $\left(\frac{4}{3} + 0i \right) \operatorname{cis} \left(\frac{\pi}{4} \right)$

$$\text{ie } \frac{4}{3} \left(\frac{1}{\sqrt{2}} + \frac{i}{\sqrt{2}} \right) = \frac{4}{3\sqrt{2}} + \frac{4i}{3\sqrt{2}}$$

$$\text{OR } \frac{2\sqrt{2}}{3} + \frac{2\sqrt{2}i}{3} \quad \checkmark$$

11 d i) let $z = x+iy$

$$z^2 + 10(z+i) = 21$$

$$(x+iy)^2 + 20x = 21$$

$$x^2 + y^2 + 20x = 21$$

$$x^2 + 20x + 100 + y^2 = 21 + 100$$

$$(x+10)^2 + y^2 = 121$$

$\therefore$ locus is circle centre $(-10, 0)$ radius 11.

ii) when $x=0$ $100 + y^2 = 121$ $\checkmark$

$$y^2 = 21$$

$$y = \pm \sqrt{21}$$

$\therefore$ numbers are $\pm i\sqrt{21}$ $\checkmark$

12.

a i) $x^2 + y^2 + z^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$
 $= (-6)^2 - 2 \times 5$
 $= 26$ ✓

ii) $\alpha^3 = -6\alpha^2 - 5\alpha + 8$ since α, β, γ are roots
 $\beta^3 = -6\beta^2 - 5\beta + 8$
 $\gamma^3 = -6\gamma^2 - 5\gamma + 8$ ✓

$\alpha^3 + \beta^3 + \gamma^3 = -6(\alpha^2 + \beta^2 + \gamma^2) - 5(\alpha + \beta + \gamma) + 24$
 $= -6 \times 26 - 5 \times (-6) + 24$
 $= -102$ ✓

iii) let $y = x^2 \Rightarrow x = \sqrt{y}$
 $y\sqrt{y} + 6y + 5\sqrt{y} - 8 = 0$
 $\sqrt{y}(y + 5) = 8 - 6y$
 $y(y + 5 + 12\sqrt{y} + 25) = 64 - 96y + 136y^2$
 $y^3 - 26y^2 + 12\sqrt{y} - 64 = 0$
 $x^3 - 26x^2 + 12x - 64 = 0$ is req'd eqn. ✓

b i) since coefficients are real $\overline{\lambda(3-i)} = 0$ ✓
 $3\lambda + 3 - i = -b$ $(3+i)(3-i) = c$
 $b = -6$ ✓ $c = 10$ ✓

ii) let $\alpha(z) = z^3 - 7z^2 + 16z - 10$
 $\alpha(1) = 1 - 7 + 16 - 10$
 $\alpha(1) = 0$
 $\therefore 1$ is a root.
 Now $z^3 - 7z^2 + 16z - 10 = (z-1)(z^2 - 6z + 10)$
 $\therefore$ roots $1, 3+i, 3-i$ ✓

c) let $P(x) = 8x^4 + 12x^3 - 30x^2 + 17x - 3$ have root mult 3.
 $P'(x) = 32x^3 + 36x^2 - 60x + 17$ " " 2
 $P''(x) = 96x^2 + 72x - 60$ " " 1
 $96x^2 + 72x - 60 = 0$
 $8x^2 + 6x - 5 = 0$
 $(4x+5)(2x-1) = 0$
 $x = \frac{1}{2}, -\frac{5}{4}$ ✓

Now $P(\frac{1}{2}) = \frac{8}{16} + \frac{12}{8} - \frac{30}{4} + \frac{17}{2} - 3$
 $= 0$

$\therefore \frac{1}{2}$ is root of mult 3.
 let roots be $\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \alpha$ ✓

sum of roots $\frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \alpha = -\frac{b}{a}$
 $\frac{3}{2} + \alpha = -\frac{3}{8}$
 $\alpha = -\frac{27}{8}$
 $\therefore \alpha = \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, -\frac{27}{8}$ ✓

B a) i) For ellipse $41p > 0$ and $91p > 0$
 $p > -4$ and $p > -9$
 $\therefore p > -4$ for ellipse ✓

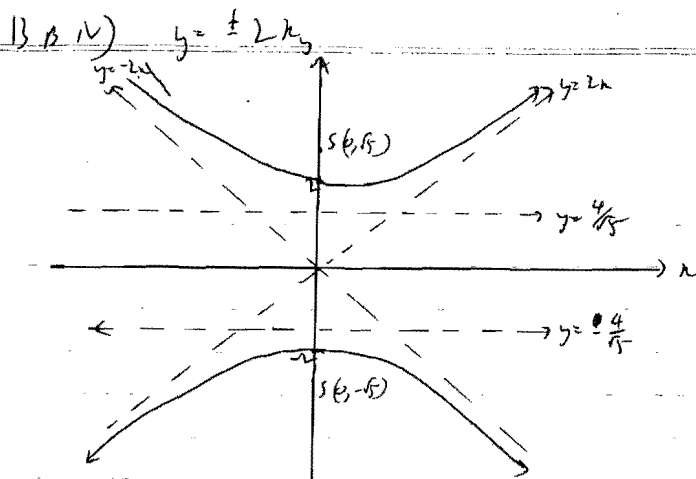
ii) For hyperbola
 $41p > 0$ and $91p < 0$ or $41p < 0$ and $91p > 0$
 $p > -4$ and $p < -9$ or $p < -4$ and $p > -9$
 impossible $-9 < p < -4$
 $\therefore$ For hyperbola $-9 < p < -4$
 (1) for $p > -9$ or (2) for $p < -4$

B) $\frac{x^2}{-1} + \frac{y^2}{4} = 1$
 $\frac{y^2}{4} - \frac{x^2}{1} = 1$

i) $a=1, b=2$
 $a^2 = 4^2(c^2 - 1)$
 $1 = 4(c^2 - 1)$
 $\frac{1}{4} = c^2 - 1$
 $c^2 = \frac{5}{4}$
 $c = \frac{\sqrt{5}}{2}$ ✓

ii) Foci $(0, \pm\sqrt{5})$ ✓

iii) Asymptotes $y = \pm \frac{2}{\sqrt{5}}x$
 $y = \pm \frac{4}{\sqrt{5}}$ or $y = \pm \frac{4\sqrt{5}}{5}$ ✓



b) diff. wrt x $2x + 8y \frac{dy}{dx} = 0$

$$\frac{dy}{dx} = -\frac{2x}{8y}$$

$$\frac{dy}{dx} = -\frac{x}{4y}$$

At $P(x_1, y_1)$ $\frac{dy}{dx} = -\frac{x_1}{4y_1}$

eqn of tangent is $y - y_1 = -\frac{x_1}{4y_1}(x - x_1)$

ii) eqn of normal is $y - y_1 = \frac{4y_1}{x_1}(x - x_1)$

At G: when $y = 0$ $-y_1 x_1 = 4y_1(x - x_1)$
 $x = -\frac{x_1}{4} + x_1 = \frac{3x_1}{4}$

$\therefore G$ is $(\frac{3x_1}{4}, 0)$

$$PG = \sqrt{\left(x_1 - \left(\frac{3x_1}{4}\right)\right)^2 + (y_1 - 0)^2}$$

$$= \sqrt{\left(\frac{x_1}{4}\right)^2 + y_1^2}$$

$$= \sqrt{\frac{x_1^2}{16} + \frac{16y_1^2}{16}}$$

$$= \sqrt{\frac{x_1^2 + 16y_1^2}{16}} = \frac{\sqrt{x_1^2 + 16y_1^2}}{4} \text{ as reqd.}$$

① if use dist or perp dist formula

13.6 iii) tangent: $y - y_1 = -\frac{x_1}{4y_1}(x - x_1)$

$$4y_1 y - 4y_1^2 = -x_1 x + x_1^2$$

$$x x_1 + 4y_1 y - (x_1^2 + 4y_1^2) = 0$$

$$x_1 x + 4y_1 y - 100 = 0$$

Perp dist O to tangent

$$PO = \frac{|0 + 0 - 100|}{\sqrt{x_1^2 + 16y_1^2}}$$

$$= \frac{100}{\sqrt{x_1^2 + 16y_1^2}}$$

$$OK = \frac{100}{x_1^2 + 16y_1^2}$$

$$PG \cdot OK = \frac{\sqrt{x_1^2 + 16y_1^2}}{4} \times \frac{100}{\sqrt{x_1^2 + 16y_1^2}}$$

$$= 25$$

$$= 5^2$$

$$\frac{x^2}{100} + \frac{y^2}{25} = 1 \text{ has semi minor axis}$$

$$PG \cdot OK = (\text{length of semi minor axis})^2$$

Q14 a) Join OA
 Let $\angle BAD = x^\circ$
 $\angle BDA = x^\circ$ (base angles of isosceles $\triangle ABD$ equal, $AB=BD$) ✓
 $\angle ODC = x^\circ$ (vertically opposite angles)
 $\angle OAD = 90^\circ - x^\circ$ (tangent $\perp$ radius at point of contact) ✓
 $\triangle OAC$ is isosceles ($OA=OC$, radii of circle)
 $\angle OCA = 90^\circ - x^\circ$ (base angles of isosceles $\triangle OCA$)
 $\angle COB = 180^\circ - (90^\circ - x^\circ) - x^\circ$ ($\angle$ sum of $\triangle COB$) ✓
 $\therefore \angle COB = 90^\circ$

b) i) $xy = c^2$
 $y = c^2 x^{-1}$
 $\frac{dy}{dx} = \frac{-c^2}{x^2}$
 At P $\frac{dy}{dx} = \frac{-c^2}{cp^2}$
 $= \frac{-1}{p^2}$

Eqn of tangent $y - \frac{c}{p} = \frac{-1}{p^2}(x - cp)$
 $py - cp = -x + cp$
 $x + py = 2cp$ (1) ✓

$m_{\text{norm}} = p^2$
 Eqn of N: $y = p^2 x$ (2) ✓
 sub (2) in (1) $x + p^2(p^2 x) = 2cp$
 $x(1 + p^4) = 2cp$
 $x = \frac{2cp}{1+p^4}$

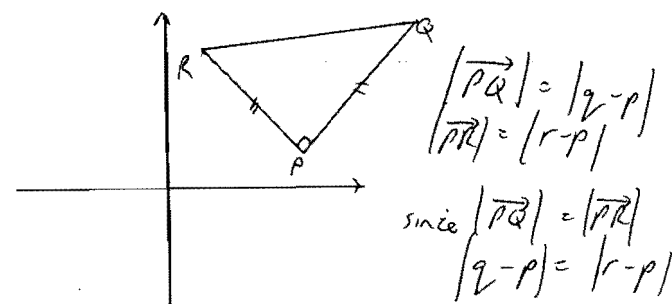
$\therefore y = p^2 \frac{2cp}{1+p^4}$
 $y = \frac{2cp^3}{1+p^4}$
 $\therefore N$ is $\left(\frac{2cp}{1+p^4}, \frac{2cp^3}{1+p^4} \right)$ ✓

14 b) ii) $y = p^2 x$
 $p^2 = \frac{y}{x}$
 $y = \frac{2cp^3}{1+p^4}$
 $y(1+p^4) = 2cp^3$
 $\left[y \left(1 + \frac{y}{x} \right) \right]^2 = \left[\frac{2cy}{x} p \right]^2$ ✓
 $y^2 \left(1 + \frac{2y}{x} + \frac{y^2}{x^2} \right) = \frac{4c^2 y^2}{x^2} \frac{y}{x}$
 $x^4 + 2xy^2 + y^4 = 4c^2 y x$
 $(x^2 + y^2)^2 = 4c^2 xy$ }

14 c) Let $P(x) = G(x)(x-a)(x-b) + R(x)$ (where $R(x) = cx + d$) ✓
 $P(a) = a^2 + d$ (1)
 $P(b) = b^2 + d$ (2)
 (1) - (2) $P(a) - P(b) = (a-b)c$
 $c = \frac{P(a) - P(b)}{a-b}$ ✓

when $c = \frac{P(a) - P(b)}{a-b}$ $P(x) = a \frac{P(a) - P(b)}{a-b} + d$
 $a \frac{P(a)}{a-b} - b \frac{P(a)}{a-b} = a \frac{P(a)}{a-b} - a \frac{P(b)}{a-b} + d$
 $d = a \frac{P(b)}{a-b} - b \frac{P(a)}{a-b}$ ✓
 $\therefore R(x) = \frac{P(a) - P(b)}{a-b} x + \frac{aP(b) - bP(a)}{a-b}$

14 d)



$$\therefore \left| \frac{q-p}{r-p} \right| = 1$$

①

✓

$$\arg(q-p) - \arg(r-p) = \pm \frac{\pi}{2}$$

$$\arg\left(\frac{q-p}{r-p}\right) = \pm \frac{\pi}{2}$$

②

✓

$$\therefore \frac{q-p}{r-p} = \operatorname{cis} \frac{\pi}{2} \quad \text{or} \quad \frac{q-p}{r-p} = \operatorname{cis} \left(-\frac{\pi}{2} \right)$$

$$\frac{q-p}{r-p} - i = 0$$

③

$$\frac{q-p}{r-p} + i = 0$$

④

✓

③ × ④

$$\left(\frac{q-p}{r-p} \right)^2 + 1 = 0$$

$$(q-p)^2 + (r-p)^2 = 0$$

$$q^2 - 2pq + p^2 + r^2 - 2pr + p^2 = 0$$

$$\therefore q^2 + r^2 + 2p^2 = 2(pq + pr)$$

✓ multiplies by
factor perf.